

ITA0609-MACHINE LEARNING

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1. A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

Answer:

Step 1: Initial Population

Population = {2, 4, 6, 8}

Fitness function:

$$f(x) = x^2$$

Individual Fitness

(x) $f(x) = x^2$

2 4

4 16

6 36

8 64

Step 2: Select the Best Candidates

The individuals with the highest fitness are:

- 8 (Fitness = 64)
- 6 (Fitness = 36)

Step 3: Binary Representation

- 8 = 1000
- 6 = 0110

Step 4: Crossover (Exchange Least Significant Bit)

Swap the last (least significant) bit:

Before crossover:

- 1000
- 0110

After swapping:

- $1000 \rightarrow 1000$ (8)
- $0110 \rightarrow 0110$ (6)

New Offspring:

- $1000 = 8$
- $0110 = 6$

(Since both LSBs are 0, the offspring remain unchanged.)

Role of Mutation

Mutation randomly changes one or more bits in a chromosome.

Example:

- $1000 \rightarrow \mathbf{1001}$ ($8 \rightarrow 9$)

Importance of Mutation:

- Increases population diversity.
- Prevents the algorithm from getting stuck in local optimum.
- Helps explore new solutions.
- Improves the chance of finding the best solution.

2. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is -0.4 . For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not.

Answer:

Given

- $w_1 = 0.5$
- $w_2 = 0.3$
- Bias $b = -0.4$
- Advertisement exposure $x_1 = 2$
- Income level $x_2 = 1$

Step 1: Compute Net Input

$$Net = w_1x_1 + w_2x_2 + b = (0.5 \times 2) + (0.3 \times 1) - 0.4 = 1.0 + 0.3 - 0.4 = 0.9$$

Step 2: Apply Step Activation Function

Step function:

$$Output = \{1, \text{ if } Net \geq 0, 0, \text{ if } Net < 0\}$$

Since:

$$0.9 \geq 0$$

Output = 1

Final Decision

- Net Input = 0.9
- Step Function Output = 1
- Prediction: The customer will purchase the product.